

Real-Time Hardware Accelerated Vision and Tracking System

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Summary

This project aims to develop a hardware accelerated, low latency object tracking system using the Nexys A7-100T FPGA. The FPGA will run an image processing system (Sobel edge detection) and a control logic (PWM), the system will achieve a high speed tracking of a physical object (e.g., a tennis ball) using 3D Printed Pan & Tilt gimbal, that will be controlled by two servo motors and it'll mount diode laser on top.

Technical Objectives

- Implement a real-time Sobel Filter pipeline in Verilog that will process raw video from a OV2640 camera sensor.
- Develop a hardware module to calculate the centroid (Center of mass) of detected edges at a target speed of 60 frames per second.
- Create a hardware PID to drive two SG90 servos motors via PWM that will be mounted on the 3D printed Pan & Tilt gimbal.
- Design a custom 3D printed mechanical gimbal to integrate the diode laser and the two servo motors into a cohesive turret.

Hardware Components

- **FPGA:** Nexys A7-100T.
- **Sensor:** OV2640 CMOS Camera.
- **Actuators:** 2x SG90 Micro Servos.
- **Pointer:** 5V 5mW Red Laser Diode driven by a PN2222 NPN transistor.
- **I/O Protocols:** I2C (Camera communication), DVP (Video data), PWM (Servo control).

Deliverables

- Verilog Source Code that will contain modular RTL design including camera interface, sobel filter, centroid calculation engine and PWM controller.
- Simulation waveforms that will contain timing diagrams of I2C configuration, pixel data alignment and PWM signal.
- Hardware utilization report containing a detailed breakdown of LUT, BRAM and DSP usage on the FPGA.
- Live demonstration showcasing the system identifying and tracking a high-contrast object (tennis ball) with a real time laser lock.